

CAŁKA NIEOZNACZONA

$$f: X_p \ni x \mapsto f(x) \in \mathbb{R}$$

$$\text{Funkcję } F: X_p \ni x \mapsto F(x) \in \mathbb{R}$$

nazywamy ^{funkcją} pierwotną (lub antyprz pochodną)

$$f(x) = x^2 \quad x \in X_p = (-\infty; +\infty)$$

$$F(x) = ?$$

$$F'(x) = f(x) \quad (\forall x \in X_p)$$

$$F(x) = \frac{1}{3}x^3$$

$$F(x) = \frac{1}{3}x^3 + 7$$

$$F(x) = \frac{1}{3}x^3 - 125$$

Całka nieoznaczona funkcji $f: X_p \rightarrow \mathbb{R}$

$$\int \underline{f(x)} dx = F(x) + C,$$

$$\text{gdzie } F'(x) = f(x)$$

IV f jest ciągła na przedziale X_p to istnieje

$$\int f(x) dx = F(x) + C$$

$$\int \frac{1}{x} dx = \ln x + C$$

$$x \in (0; +\infty)$$

e^{x^2} jest ciągła na przedziale X_p

$$\int e^{x^2} dx$$
$$\int \sin x^2 dx$$

Własności całki

↳ całka jest funkcją liniową, czyli:

$$(1) \int (f_1 \pm f_2) dx = \int f_1 dx \pm \int f_2 dx$$

$$(2) \int a \cdot f dx = a \int f dx$$

Wzory całkowe

Wzór dla humanistów

$$\int x^n dx = \begin{cases} \frac{1}{n+1} \cdot x^{n+1}, & \text{dla } n \in \mathbb{R} \setminus \{-1\} \\ \int x^{-1} dx = \int \frac{1}{x} dx = \ln x \end{cases}$$

$$\int \cos x dx = \sin x$$

$$\int \frac{1}{\sin^2 x} dx = -\operatorname{ctg} x$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x$$

$$\int \sin x dx = -\cos x$$

$$\int e^x dx = e^x$$

$$\int \frac{1}{\cos^2 x} dx = \operatorname{tg} x$$

$$\int \frac{1}{1+x^2} dx = \operatorname{arctg} x$$

$$\int \frac{f'(x)}{f(x)} dx = \ln f(x) \quad \text{WARTO ZAPAMIĘTAĆ}$$

$$\text{np. } (\ln f(x))' = \frac{1}{f(x)} \cdot f'(x) = \frac{f'(x)}{f(x)}$$

Przykład

$$\begin{aligned} & \int (3x^2 - 4x^{\frac{1}{3}} - \frac{2}{x} + \frac{5}{1+x^2} - 6e^x + 8\cos x - 2\sin x + \frac{x}{x^2+1}) dx = \\ &= 3 \int x^2 dx - 4 \int x^{\frac{1}{3}} dx - 2 \int \frac{1}{x} dx + 5 \int \frac{1}{1+x^2} dx - 6 \int e^x dx + \\ &+ 8 \int \cos x dx - 2 \int \sin x dx + \int \frac{x}{x^2+1} dx = \\ &= 3 \cdot \frac{1}{3} x^3 - 4 \frac{1}{1+\frac{1}{3}} x^{\frac{4}{3}} - 2 \ln x + 5 \arctg x - 6e^x + \\ &+ 8 \sin x + 2 \cos x + \frac{1}{2} \cdot \ln(x^2+1) + C \end{aligned}$$

$$\int \frac{1}{x \ln x} dx = \int \frac{\frac{1}{x}}{\ln x} dx = \ln(\ln x) + C$$

$$\int dx = \int 1 dx$$

$$\int \tg x dx = \int \frac{\sin x}{\cos x} dx = -1 \cdot \int \frac{-\sin x}{\cos x} dx = -\ln(\cos x) + C$$

$$\begin{aligned} \int \ctg^2 x dx &= \int \frac{\cos^2 x}{\sin^2 x} dx = \int \frac{1 - \sin^2 x}{\sin^2 x} dx = \\ &= \int \frac{1}{\sin^2 x} dx - \int 1 dx = -\ctg x - x + C \end{aligned}$$

$$\int \frac{x^2+1-1}{1+x^2} dx = \int \left(1 - \frac{1}{x^2+1}\right) dx =$$

$$= \int 1 dx - \int \frac{1}{x^2+1} dx = x - \arctg x + C$$

$$\int \frac{\sin^2 x}{1+\cos x} dx = \int \frac{1-\cos^2 x}{1+\cos x} dx = \int (1-\cos x) dx =$$

$$= \int 1 dx - \int \cos x dx = x - \sin x + C$$

Całkowanie przez części

$$\int f \cdot g' dx = f \cdot g - \int f' \cdot g dx$$

d-d

$$(f \cdot g)' = f'g + f \cdot g' \quad | \int () dx$$

$$\int (f \cdot g)' dx = \int f'g dx + \int f \cdot g' dx$$

$$fg = \int f'g dx + \int f \cdot g' dx$$

$$\int f \cdot g' dx = fg - \int f'g dx$$

Dobre rady

→ gdy iloczyn dwóch funkcji

→ gdy jest logarytm to często się stosuje

$$\int \ln x \, dx = \int 1 \cdot \ln x \, dx = \left\{ \begin{array}{l} f = \ln x \quad f' = \frac{1}{x} \\ g' = 1 \quad g = x \end{array} \right\} =$$

$$= x \ln x - \int \frac{1}{x} \cdot x \, dx = x \ln x - \int dx = x \ln x - x + C$$

$$\int \underset{\substack{\uparrow \\ g'}}{1} \cdot \underset{\substack{\uparrow \\ f}}{\ln^2 x} \, dx = \left\{ \begin{array}{l} f = \ln^2 x \quad f' = 2 \ln x \cdot \frac{1}{x} \\ g' = 1 \quad g = x \end{array} \right\} =$$

$$= \ln^2 x \cdot x - \int x \cdot 2 \ln x \cdot \frac{1}{x} \, dx = \ln^2 x \cdot x - 2 \int \ln x \, dx =$$

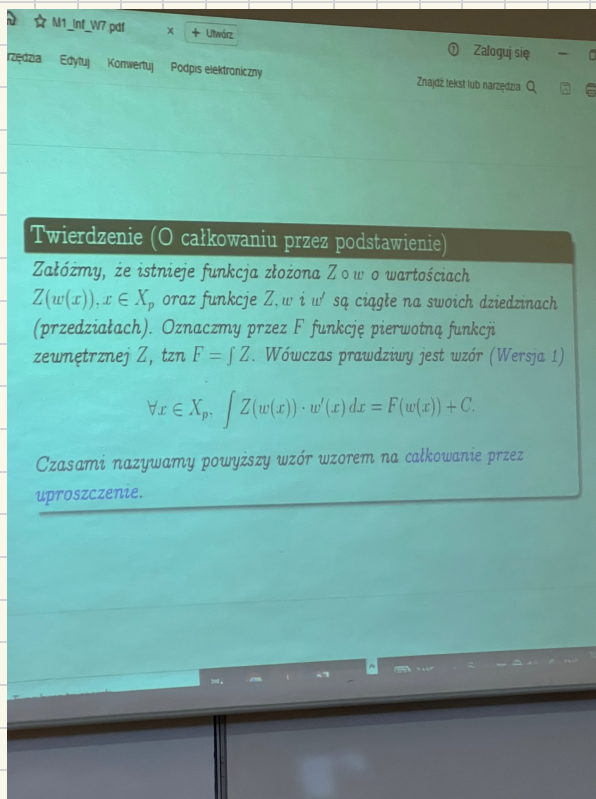
$$= \ln^2 x \cdot x - 2 \int 1 \cdot \ln x \, dx = \left\{ \begin{array}{l} f = \ln x \quad f' = \frac{1}{x} \\ g' = 1 \quad g = x \end{array} \right\} =$$

$$= \ln^2 x \cdot x - 2 \int x \cdot \frac{1}{x} \, dx = \ln^2 x \cdot x - 2 \ln x \cdot x - 2x + C$$

$$\int x \cdot e^x \, dx = \left\{ \begin{array}{l} f = e^x \quad f' = e^x \\ g' = x \quad g = \frac{1}{2} x^2 \end{array} \right\} = \frac{1}{2} x^2 \cdot e^x - \int \frac{1}{2} x^2 \cdot e^x \, dx \quad \text{steig. macht}$$

$$\int x \cdot e^x \, dx = \left\{ \begin{array}{l} f = x \quad f' = 1 \\ g' = e^x \quad g = e^x \end{array} \right\} = x \cdot e^x - \int e^x \, dx =$$

$$= x \cdot e^x - e^x + C$$



inna metoda
na całkowanie
po części

Całkowanie przez podstawienie

$$\int (x+7)^{15} dx = \left\{ \begin{array}{l} x+7=t \\ 1 dx = 1 dt \\ (x+7)^{15} dx = t^{15} dt \end{array} \right\} = \int t^{15} dt = \frac{1}{16} \cdot t^{16} = \frac{1}{16} (x+7)^{16} + C$$

$$\int \frac{x^3}{\sqrt{1-4x^4}} dx = \left\{ \begin{array}{l} 1-4x^4 = t \\ -16x^3 dx = 1 dt \\ \frac{x^3}{\sqrt{1-4x^4}} dx = \frac{x^3 \cdot dx}{\sqrt{t}} = \frac{-dt}{\frac{1}{16}\sqrt{t}} \end{array} \right\} =$$

$$= -\frac{1}{16} \int \frac{1}{\sqrt{t}} dt = -\frac{1}{16} \int t^{-\frac{1}{2}} dt =$$

$$= -\frac{1}{16} \left(\frac{1}{1-\frac{1}{2}} \cdot t^{1-\frac{1}{2}} \right) = -\frac{1}{8} \sqrt{t} = -\frac{1}{8} \sqrt{1-4x^4} + C$$